HACKTOBERFEST HACK DAY | THEME 01

OPEN MODEL EVALUATION

A hands-on comparison of four open-weight models across a 25-prompt test plan.



4	25	100	13
models selected	standard prompts	planned runs	scored runs in current work log

Evidence-based draft report and presentation • 24 October 2026

Objective and evaluation design

Question we are trying to answer

Which open-weight model is most useful for a target application when correctness, instruction following, clarity and honesty are judged on the same prompts?

Planned design: 25 prompts × 4 models = 100 runs

Five categories: reasoning/math, factual accuracy, instruction following, coding, and Indian languages.

MODEL SET

gpt-oss-20b	Groq Playground
gpt-oss-120b	Groq Playground
Qwen 3.8 27B (Preview)	Groq Playground
Gemma 4 31B	Google AI Studio

Current evidence: 13 scored runs; all are reasoning/math. The study is therefore preliminary.

Model factsheet at a glance

Model	Size / architecture	Context	Licence & documented strengths
gpt-oss-20b	20.91B total; 3.61B active MoE	128K hosted	Apache 2.0 + usage policy Reasoning, coding, tools
gpt-oss-120b	116.83B total; 5.13B active MoE	128K hosted	Apache 2.0 + usage policy Higher-capability reasoning, tools
Qwen 3.8 27B	27B dense Hybrid attention	262K native; Groq 131K	Apache 2.0 Coding, vision, agentic workflows
Gemma 4 31B	30.7B dense	256K documented	Apache 2.0 Reasoning, coding, image input; broad family multilingual support

Native model context and hosted endpoint limit differ: report both; verify the current UI setting used in your run.

Evaluation protocol and scoring rubric

01 Same prompt
Use identical wording for all models.

02 Same settings
Same system instruction and temperature; record differences.

03 Fresh chat
Start a new conversation per prompt.

04 Two reviewers
Score independently, then agree a final score.

05 Evidence
Keep response text and screenshots for unusual results.

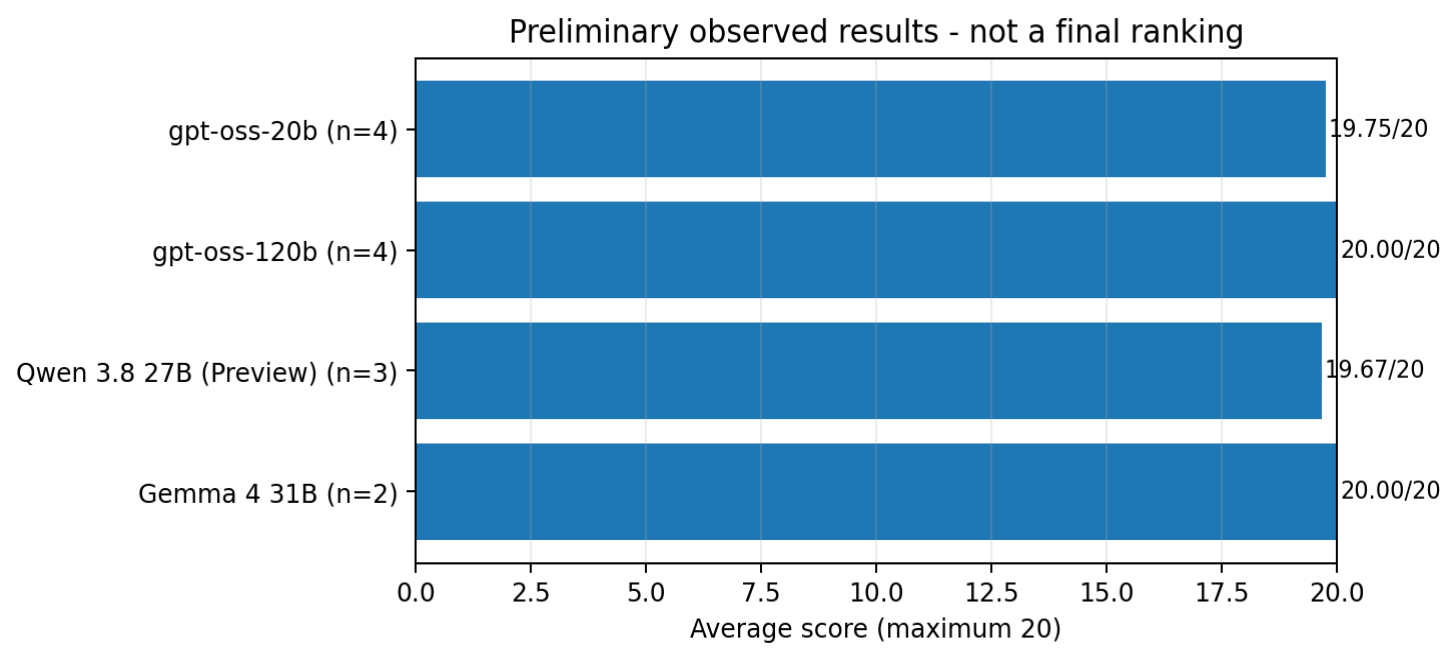
4 criteria × 1–5 points = 20 total

Correctness • Instruction following • Clarity • Honesty
1 = weak, 3 = partial/acceptable, 5 = fully meets criteria.

Important setting audit

The actual temperature and account-session settings were not available in the evidence received here. Confirm them from Groq and AI Studio before claiming a fully controlled run.

What the recorded scores show



258 / 260 points

99.23% of available points in the partial sample

13 of 100 runs

Only reasoning/math runs are scored here

Per-model summary

gpt-oss-120b • 20.00/20 (n=4) | Gemma 4 31B • 20.00/20 (n=2)
gpt-oss-20b • 19.75/20 (n=4) | Qwen 3.8 27B • 19.67/20 (n=3)

Do not compare these averages as a definitive ranking: sample sizes differ and the test items are simple.

Findings: strengths, caveats and gaps

Observed in the recorded responses

- Prompt 1: all four answered ₹5 correctly.
- Prompt 4: all four gave 210 km.
- Prompt 5: both identified gpt-oss models answered 9 sheep.
- Prompt 2: two models had a minor clarity/formatting deduction.

What we cannot conclude yet

- No empirical coding winner.
- No scored evidence for fact checking or Indian languages.
- No safety or summarisation category in the current bank.
- No complete second-scorer consensus documented.

Failure pattern so far: not enough failures to make a defensible pattern.

These questions are easy and produce ceiling scores. A good model evaluation needs harder edge cases, contradictory premises, strict format constraints, executable code tests, and fluent-speaker review for language quality.

Report the missing data openly; never fill blank score cells with estimates.

Application recommendations — provisional

STUDENT HELP-DESK

Pilot gpt-oss-120b + gpt-oss-20b

Use the same campus FAQ test set. Require source-grounded answers, uncertainty, and escalation when facts are missing. Add safety and privacy checks.

CODING HELPER

Benchmark Qwen 3.8 27B + gpt-oss-120b

Official documentation points to coding/agentic uses, but the current workbook has no coding scores. Execute code and test edge cases before choosing.

INDIAN-LANGUAGE APP

Pilot Gemma 4 31B + Qwen 3.8 27B

Documentation reports broad multilingual capability. Test Hindi and a second target language with fluent human reviewers; broad model claims do not guarantee idiomatic output.

Before final recommendation: finish the full prompt set, record latency/settings, and obtain two-scorer agreement.

Live demo: compare two models in real time

“Write a Python function that returns the second-largest distinct number in a list. Explain how it handles duplicates.”

1 Ask

Invite the audience to approve the prompt.

2 Run

Use new chats on gpt-oss-120b and Qwen 3.8 27B with matched settings.

3 Test

Try duplicates, negatives and fewer than two distinct values.

4 Score

Two reviewers score correctness, instructions, clarity and honesty.

5 Discuss

Explain any trade-off; do not generalise from one prompt.

Show the model ID and temperature. Keep both outputs visible. Agree on a score after independent ratings.

Path to a defensible final result

Complete coverage	25 prompts × 4 models = 100 runs; 13 scored runs currently documented.
Record settings	System prompt, temperature, exact model IDs, date and any platform differences.
Double-score	Independent reviewer scores, discussion and final agreed score for every response.
Capture evidence	Screenshots/full outputs for failures, formatting issues and notable model differences.
Refresh analysis	Calculate averages by model and category only after coverage is complete.

Conclusion today: all recorded models handle the simple reasoning items well; the full evaluation is needed to identify task-specific winners.

Public sources used

Project instructions & worksheet	github.com/ankit2738/HacktoberFest-HackDay/.../01_model_evaluation.md
OpenAI gpt-oss launch & model card	openai.com/index/introducing-gpt-oss/ • deploymentsafety.openai.com/gpt-oss
Groq model pages	console.groq.com/docs/model/openai/gpt-oss-20b • .../gpt-oss-120b • .../qwen/qwen3.8-27b
Qwen official model repo / announcement	github.com/AlibabaCloud-Official/Qwen3.8-27B • alibabacloud.com/blog/alibaba-unveils-qwen3-8-27b_603463
Google Gemma 4 model card / release notes	ai.google.dev/gemma/docs/core/model_card_4 • ai.google.dev/gemma/docs/releases

Thank you | Questions? | Suggest a prompt for the live comparison.